

✓ Welcome to Colab!

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
!pip install openpyxl
```

Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_excel('Week 4.xlsx')
```

```
df.head()
```

1 to 5 of 5 entries Filter ?

index	Id	ActivityDate	TotalSteps	Calories	AverageHeartRate	SleepHours	Unnamed: 6 ▲	Unnamed: 7
0	1503960366	3/25/2016	11004	1819	85	7.5	NaN	NaN
1	1503960366	3/26/2016	17609	2154	95	7.5	NaN	NaN
2	1503960366	3/27/2016	12736	1944	95	7.5	NaN	NaN
3	1503960366	3/28/2016	13231	1932	95	7.5	NaN	NaN
4	1503960366	3/29/2016	12041	1886	95	7.5	NaN	NaN

Show per page

Like what you see? Visit the [data table notebook](#) to learn more about interactive tables.

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 457 entries, 0 to 456
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Id                    457 non-null   int64
1   ActivityDate          457 non-null   object
2   TotalSteps            457 non-null   int64
3   Calories              457 non-null   int64
4   AverageHeartRate      457 non-null   int64
5   SleepHours            457 non-null   float64
6   Unnamed: 6            0 non-null     float64
7   Unnamed: 7            1 non-null     object
dtypes: float64(2), int64(4), object(2)
memory usage: 28.7+ KB
```

```
df.columns
```

```
Index(['Id', 'ActivityDate', 'TotalSteps', 'Calories', 'AverageHeartRate',
       'SleepHours', 'Unnamed: 6', 'Unnamed: 7'],
      dtype='object')
```

```
df['TotalSteps'].mean()
```

```
np.float64(6546.562363238512)
```

```
df['Calories'].mean()
```

```
np.float64(2189.45295404814)
```

```
df['AverageHeartRate'].mean()
```

```
np.float64(80.63676148796499)
```

```
df['SleepHours'].mean()

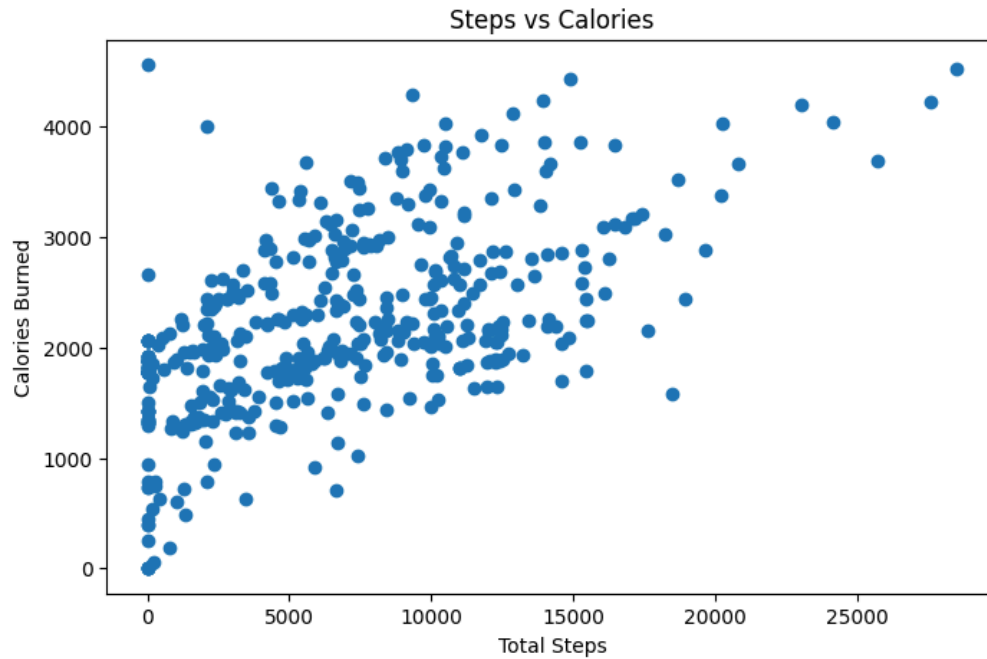
np.float64(7.4866520787746165)
```

```
plt.figure(figsize=(8,5))

plt.scatter(df['TotalSteps'], df['Calories'])

plt.xlabel('Total Steps')
plt.ylabel('Calories Burned')
plt.title('Steps vs Calories')

plt.show()
```



```
df.corr(numeric_only=True)
```

	Id	TotalSteps	Calories	AverageHeartRate	SleepHours	Unnamed: 6	
Id	1.000000	0.138662	0.290868	0.117769	-0.287845	NaN	
TotalSteps	0.138662	1.000000	0.581380	0.939800	-0.551602	NaN	
Calories	0.290868	0.581380	1.000000	0.529309	-0.914281	NaN	
AverageHeartRate	0.117769	0.939800	0.529309	1.000000	-0.516749	NaN	
SleepHours	-0.287845	-0.551602	-0.914281	-0.516749	1.000000	NaN	
Unnamed: 6	NaN	NaN	NaN	NaN	NaN	NaN	

```
top_users = df.groupby('Id')['TotalSteps'].mean().sort_values(ascending=False)

top_users.head(10)
```

TotalSteps	
Id	
8877689391	17417.083333
8053475328	14844.363636
6962181067	12639.714286
7007744171	12260.333333
2022484408	12174.916667
1503960366	11640.526316
2347167796	9800.066667
1644430081	9274.800000
3977333714	8663.916667
5577150313	8608.181818

dtype: float64

```
df.to_excel('final_fitness_project.xlsx', index=False)
```



Start coding or [generate](#) with AI.